

Sequences and Series (Part III)

Date
Course

Sigma Notation

If "a" and "b" are positive integers (i.e. positive whole numbers), then

$$\sum_{n=a}^b f(n) = f(a) + f(a+1) + f(a+2) + \dots + f(b)$$

Last term $\downarrow$

where $f(n)$ is an algebraic expression in terms of "n," but other variables such as "k," "r," "m," or "p" are often used in place of "n."

Find each sum.

• ① $\sum_{n=1}^5 3^n = 3^1 + 3^2 + 3^3 + 3^4 + 3^5 = 3 + 9 + 27 + 81 + 243 = \boxed{363}$

② $\sum_{n=0}^3 2n = 2(0) + 2(1) + 2(2) + 2(3) = 0 + 2 + 4 + 6 = \boxed{12}$

③ $\sum_{n=1}^3 4\left(-\frac{1}{2}\right)^{n-1} = 4\left(-\frac{1}{2}\right)^{1-1} + 4\left(-\frac{1}{2}\right)^{2-1} + 4\left(-\frac{1}{2}\right)^{3-1} = 4\left(-\frac{1}{2}\right)^0 + 4\left(-\frac{1}{2}\right)^1 + 4\left(-\frac{1}{2}\right)^2$
 $= 4(1) + 4\left(-\frac{1}{2}\right) + 4\left(\frac{1}{4}\right)$
 $= 4 - 2 + 1$
 $= \boxed{3}$

④ $\sum_{n=2}^5 n(n+4) = 2(2+4) + 3(3+4) + 4(4+4) + 5(5+4)$
 $= 2(6) + 3(7) + 4(8) + 5(9)$
 $= 12 + 21 + 32 + 45$
 $= \boxed{110}$

• ⑤ $\sum_{n=1}^4 n-3 = (1-3) + (2-3) + (3-3) + (4-3) = -2 - 1 + 0 + 1$
 $= \boxed{-2}$

⑥ $\sum_{n=2}^6 n^2 = 2^2 + 3^2 + 4^2 + 5^2 + 6^2 = 4 + 9 + 16 + 25 + 36 = \boxed{90}$

⑦ $\sum_{n=1}^3 \frac{1}{3}(2)^{n-1} = \frac{1}{3}(2)^{1-1} + \frac{1}{3}(2)^{2-1} + \frac{1}{3}(2)^{3-1} = \frac{1}{3}(2)^0 + \frac{1}{3}(2)^1 + \frac{1}{3}(2)^2$
 $= \frac{1}{3}(1) + \frac{1}{3}(2) + \frac{1}{3}(4)$
 $= \frac{1}{3} + \frac{2}{3} + \frac{4}{3}$
 $= \boxed{\frac{7}{3}}$

⑧ $\sum_{n=0}^3 7 - \frac{1}{4}n = 7 - \frac{1}{4}(0) + 7 - \frac{1}{4}(1) + 7 - \frac{1}{4}(2) + 7 - \frac{1}{4}(3)$
 $= 7 - 0 + 7 - \frac{1}{4} + 7 - \frac{1}{2} + 7 - \frac{3}{4}$
 $= 4(7) - \frac{6}{4}$
 $= \boxed{26\frac{1}{2}}$

Determine if each series is arithmetic or geometric.
Then find the sum.

$$\begin{aligned} \textcircled{9} \sum_{n=1}^{100} 3n+4 &= (3(1)+4) + (3(2)+4) + (3(3)+4) + (3(4)+4) + \dots + (3(100)+4) \\ &= 7 + 10 + 13 + 16 + \dots + 304 \\ &\quad \underbrace{\quad \quad \quad}_{+3} \underbrace{\quad \quad \quad}_{+3} \underbrace{\quad \quad \quad}_{+3} \end{aligned}$$

Arithmetic

$$S_n = \frac{n}{2}(2a_1 + D(n-1))$$

$$\begin{aligned} S_{100} &= \frac{100}{2}(2(7) + 3(100-1)) \\ &= 50(14 + 3(99)) \\ &= 50(14 + 297) \\ &= 50(311) \\ &= \boxed{15,550} \end{aligned}$$

$$\textcircled{10} \sum_{n=1}^{54} \left(\frac{1}{5}\right)^n = \left(\frac{1}{5}\right)^1 + \left(\frac{1}{5}\right)^2 + \left(\frac{1}{5}\right)^3 + \left(\frac{1}{5}\right)^4 + \dots + \left(\frac{1}{5}\right)^{54}$$

$$= \frac{1}{5} + \frac{1}{25} + \frac{1}{125} + \frac{1}{625} + \dots + \left(\frac{1}{5}\right)^{54}$$

$\underbrace{\quad \quad \quad}_{\times \frac{1}{5}} \quad \underbrace{\quad \quad \quad}_{\times \frac{1}{5}} \quad \underbrace{\quad \quad \quad}_{\times \frac{1}{5}}$

Geometric

$$S_n = a_1 \frac{1-r^n}{1-r}$$

$$S_{54} = \left(\frac{1}{5}\right) \frac{1-\left(\frac{1}{5}\right)^{54}}{1-\frac{1}{5}}$$

$$S_{54} = \left(\frac{1}{5}\right) \frac{1-\left(\frac{1}{5}\right)^{54}}{\frac{4}{5}}$$

$$= \frac{1}{5} \frac{\left(1-\left(\frac{1}{5}\right)^{54}\right) 5}{\left(\frac{4}{5}\right) 5}$$

$$= \boxed{\frac{1-\left(\frac{1}{5}\right)^{54}}{4}}$$

$$\textcircled{11} \sum_{n=1}^{27} -\frac{n}{7} + 10 = -\frac{1}{7} + 10 - \frac{2}{7} + 10 - \frac{3}{7} + 10 - \frac{4}{7} + 10 + \dots + (-\frac{27}{7} + 10)$$

$$= 9\frac{6}{7} + 9\frac{5}{7} + 9\frac{4}{7} + 9\frac{3}{7} + \dots + (-\frac{27}{7} + 10)$$

Arithmetic

$$S_n = \frac{n}{2}(2a_1 + D(n-1))$$

$$S_{27} = \frac{27}{2}(2(9\frac{6}{7}) - \frac{1}{7}(27-1))$$

$$= \frac{27}{2}(18 + \frac{12}{7} - \frac{1}{7}(26))$$

$$= \frac{27}{2}(18 - \frac{14}{7})$$

$$= \frac{27}{2}(18-2)$$

$$= \frac{27}{2}(16)$$

$$= 27(8)$$

$$= \boxed{216}$$

$$\textcircled{12} \sum_{n=1}^{45} 5(-1.5)^{n-1} = 5(-1.5)^{1-1} + 5(-1.5)^{2-1} + 5(-1.5)^{3-1} + \dots + 5(-1.5)^{45-1}$$

$$= 5(-1.5)^0 + 5(-1.5)^1 + 5(-1.5)^2 + \dots + 5(-1.5)^{44}$$

$$= 5(1) + 5(-1.5) + 5(2.25) + \dots + 5(-1.5)^{44}$$

$$= 5 - 7.5 + 11.25 + \dots + 5(-1.5)^{44}$$

Geometric

$$S_n = a_1 \frac{1-r^n}{1-r}$$

$$S_{45} = (5) \frac{1-(-1.5)^{45}}{1-(-1.5)}$$

$$= 5 \frac{1+1.5^{45}}{2.5}$$

$$= \frac{5}{2.5}(1+1.5^{45})$$

$$= \boxed{2(1+1.5^{45})}$$

• ⑬ $\sum_{n=1}^{31} (-2)^{n+1} = 4 - 8 + 16 - 32 + \dots + (-2)^{31+1}$

Geometric

$$S_n = a_1 \frac{1-r^n}{1-r}$$

$$S_{31} = 4 \frac{1-(-2)^{31}}{1-(-2)}$$

$$= 4 \frac{(1+2^{31})}{1+2}$$

$$= \boxed{\frac{4}{3}(1+2^{31})}$$

⑭ $\sum_{n=1}^{40} 2+n = 3 + 4 + 5 + 6 + \dots + 2+40$

Arithmetic

$$S_n = \frac{n}{2}(2a_1 + d(n-1))$$

$$S_{40} = \frac{40}{2}(2(3) + (1)(40-1))$$

$$= 20(6 + (1)(39))$$

$$= 20(45)$$

$$= \boxed{900}$$

$$\textcircled{15} \sum_{n=1}^{10} 2(0.6)^n = 2(0.6)^1 + 2(0.6)^2 + 2(0.6)^3 + 2(0.6)^4 + \dots + 2(0.6)^{10}$$

$$= 1.2 + 0.72 + 0.432 + 0.2592 + \dots + 2(0.6)^{10}$$

Geometric

$$S_n = a_1 \frac{1-r^n}{1-r}$$

$$S_{10} = (1.2) \frac{1-(0.6)^{10}}{1-0.6}$$

$$= (1.2) \frac{1-0.6^{10}}{0.4}$$

$$= \frac{1.2}{0.4} (1-0.6^{10})$$

$$= \boxed{3(1-0.6^{10})}$$

$$\textcircled{16} \sum_{n=1}^{65} -2n-9 = -11-13-15-17+\dots+(-2(65)-9)$$

Arithmetic

$$S_n = \frac{n}{2}(2a_1 + D(n-1))$$

$$S_{65} = \frac{65}{2}(2(-11) - 2(65-1))$$

$$= \frac{65}{2}(-22 - 2(64))$$

$$= \frac{65}{2}(-150)$$

$$= 65(-75)$$

$$= \boxed{-4,875}$$

$$\bullet \textcircled{17} \sum_{n=0}^{20} \frac{1}{3000}(3)^n = \frac{1}{3000} + \frac{3}{3000} + \frac{9}{3000} + \dots + \frac{1}{3000}(3)^{20}$$

Geometric

$$S_n = a_1 \frac{1-r^n}{1-r}$$

$$S_{21} = \frac{1}{3000} \frac{1-3^{21}}{1-3}$$

$$= \frac{1}{3000} \frac{1-3^{21}}{-2}$$

0, 1, 2, 3, 4 ~ 20
n=21

$$\boxed{-\frac{(1-3^{21})}{6000}}$$

$$(18) \sum_{n=2}^{34} -4n+5 = -3 -7 -11 -15 + \dots + (-4(34)+5)$$

Arithmetic

$$S_n = \frac{n}{2}(2a_1 + D(n-1))$$

$$S_{33} = \frac{33}{2}(2(-3) - 4(33-1))$$

$$= \frac{33}{2}(-6 - 128)$$

$$= \frac{33}{2}(-134)$$

$$= 33(-67)$$

$$= \boxed{-2,211}$$

$$(19) \sum_{n=3}^{28} 3n = 9 + 12 + 15 + \dots + 3(28)$$

Arithmetic

$$S_n = \frac{n}{2}(2a_1 + D(n-1))$$

$$S_{26} = \frac{26}{2}(2(9) + 3(26-1))$$

$$= 13(18 + 3(25))$$

$$= 13(93)$$

$$= \boxed{1,209}$$

~~1, 2, 3, 4, 5~~ 28
n=26

$$\textcircled{20} \sum_{n=0}^{16} (-0.9)^{n+1} = -0.9 + 0.81 - 0.729 + \dots + (-0.9)^{16+1}$$

Geometric

$$S_n = a_1 \frac{1-r^n}{1-r}$$

$$S_n = (-0.9) \frac{1-(-0.9)^{17}}{1-(-0.9)}$$

$$= -0.9 \frac{1+(0.9)^{17}}{1.9}$$

$$= \frac{-0.9}{1.9} (1+0.9^{17})$$

$$= \boxed{-\frac{9}{19} (1+0.9^{17})}$$

0, 1, 2, 3... 16

$n=17$

$$\bullet \textcircled{21} \sum_{n=4}^{300} \frac{n}{5} = \frac{4}{5} + \frac{5}{5} + \frac{6}{5} + \frac{7}{5} + \dots + \frac{300}{5}$$

Arithmetic

$$S_n = \frac{n}{2} (2a_1 + D(n-1))$$

$$S_{297} = \frac{297}{2} (2(\frac{4}{5}) + \frac{1}{5}(297-1))$$

$$= \frac{297}{2} (\frac{8}{5} + \frac{296}{5})$$

$$= \frac{297}{2} (\frac{304}{5})$$

$$= \frac{90,288}{10}$$

$$= \boxed{9,028.8}$$

$$\textcircled{22} \sum_{n=2}^{49} (-2)^{n-1} = -2 + 4 - 8 + 16 + \dots + (-2)^{49-1}$$

Geometric

$$S_n = a_1 \frac{1-r^n}{1-r}$$

$$S_{48} = (-2) \frac{1-(-2)^{48}}{1-(-2)}$$

$$= (-2) \frac{1-2^{48}}{3}$$

$$= \boxed{-\frac{2}{3}(1-2^{48})}$$

$$\textcircled{23} \sum_{n=0}^{21} -6n+10 = 10+4-2-8+\dots-6(21)+10$$

Arithmetic

$$S_n = \frac{n}{2}(2a_1 + D(n-1))$$

$$S_{22} = \frac{22}{2}(2(10) - 6(22-1))$$

$$= 11(20 - 126)$$

$$= 11(-106)$$

$$= \boxed{-1,166}$$

$$\textcircled{24} \sum_{n=3}^{44} 2\left(\frac{1}{5}\right)^n = \frac{2}{125} + \frac{2}{625} + \frac{2}{3125} + \dots + 2\left(\frac{1}{5}\right)^{44}$$

Geometric

$$S_n = a_1 \frac{1-r^n}{1-r}$$

$$S_{42} = \frac{2}{125} \frac{1-\left(\frac{1}{5}\right)^{42}}{1-\frac{1}{5}}$$

$$\frac{2}{125} \frac{1-\left(\frac{1}{5}\right)^{42}}{\frac{4}{5}}$$

$$\frac{2}{12} \left(1-\left(\frac{1}{5}\right)^{42}\right)$$

$$\frac{1}{6} \left(1-\left(\frac{1}{5}\right)^{42}\right)$$

$$\boxed{\frac{1-\left(\frac{1}{5}\right)^{42}}{6}}$$

$$\bullet \textcircled{25} \sum_{n=1}^{24} \frac{1}{2}n - \frac{1}{2} = 0 + \frac{1}{2} + 1 + 1\frac{1}{2} + \dots + \frac{1}{2}(24) - \frac{1}{2}$$

Arithmetic

$$S_n = \frac{n}{2}(2a_1 + D(n-1))$$

If $a_1 = 0$, then $n = 24$

If $a_1 = \frac{1}{2}$, then $n = 23$

$$S_{24} = \frac{24}{2} \left(2(0) + \frac{1}{2}(24-1)\right) \quad S_{23} = \frac{23}{2} \left(2\left(\frac{1}{2}\right) + \frac{1}{2}(23-1)\right)$$

$$= 12 \left(0 + \frac{23}{2}\right)$$

$$= \boxed{138}$$

$$= \frac{23}{2} (1+11)$$

$$= \frac{23}{2} (12)$$

$$= \boxed{138}$$

$$\textcircled{26} \sum_{n=1}^{67} -n+1 = 0 - 1 - 2 - 3 + \dots + (-67+1)$$

Arithmetic

$$S_n = \frac{n}{2}(2a_1 + D(n-1))$$

$$S_{67} = \frac{67}{2}(2(0) - 1(67-1))$$

$$= \frac{67}{2}(0 - 66)$$

$$= \frac{67}{2}(-66)$$

$$= \boxed{-2,211}$$

$$\bullet \textcircled{27} \sum_{n=1}^{38} 3(n-1) = 0 + 3 + 6 + 9 + \dots + 3(38-1)$$

Arithmetic

$$S_n = \frac{n}{2}(2a_1 + D(n-1))$$

$$S_{38} = \frac{38}{2}(2(0) + 3(38-1))$$

$$= 19(0 + 111)$$

$$= \boxed{2,109}$$

$$\textcircled{28} \sum_{n=1}^{18} \frac{5n-1}{2} - 2 = 0 + \left(\frac{9}{2} - 2\right) + 5 + \frac{19}{2} - 2 + \dots + \frac{5(18)-1}{2} - 2$$

$$= 0 + \frac{5}{2} + 5 + \frac{15}{2} + \dots + \frac{89}{2} - 2$$

$$= 0 + 2.5 + 5 + 7.5 + \dots + 42.5$$

Arithmetic

$$S_n = \frac{n}{2}(2a_1 + D(n-1))$$

$$S_{18} = \frac{18}{2}(2(0) + 2.5(18-1))$$

$$= 9(0 + 2.5(17))$$

$$= 9(42.5)$$

$$= \boxed{382.5}$$

$$\bullet \textcircled{29} \sum_{n=2}^{80} n-2 = 0 + 1 + 2 + 3 + \dots + 80-2$$

Arithmetic

$$S_n = \frac{n}{2}(2a_1 + D(n-1))$$

$$S_{79} = \frac{79}{2}(2(0) + 1(79-1))$$

$$= \frac{79}{2}(0 + 78)$$

$$= \boxed{3,081}$$

$$\textcircled{30} \sum_{n=0}^{75} \frac{n}{3} = 0 + \frac{1}{3} + \frac{2}{3} + \frac{3}{3} + \dots + \frac{75}{3}$$

Arithmetic

$$S_n = \frac{n}{2}(2a_1 + D(n-1))$$

$$S_{76} = \frac{76}{2}(2(0) + \frac{1}{3}(76-1))$$

$$= 38(0 + 25)$$

$$= \boxed{950}$$

$$\textcircled{31} \sum_{n=3}^{22} 6-2n = 0 - 2 - 4 - 6 + \dots + 6 - 2(22)$$

Arithmetic

$$S_n = \frac{n}{2}(2a_1 + D(n-1))$$

$$S_{20} = \frac{20}{2}(2(0) - 2(20-1))$$

$$= 10(0 - 38)$$

$$= \boxed{-380}$$

• (32) $\sum_{n=0}^{13} 4n = 0 + 4 + 8 + 12 + \dots + 4(13)$

Arithmetic

$$S_n = \frac{n}{2}(2a_1 + D(n-1))$$

$$S_{14} = \frac{14}{2}(2(0) + 4(14-1))$$

$$= 7(0 + 52)$$

$$= \boxed{364}$$

(33) $\sum_{n=2}^{17} \frac{2n}{3} - \frac{4}{3} = 0 + \frac{2}{3} + \frac{4}{3} + \frac{6}{3} + \frac{8}{3} + \dots + \frac{2(17)}{3} - \frac{4}{3}$

Arithmetic

$$S_n = \frac{n}{2}(2a_1 + D(n-1))$$

$$S_{16} = \frac{16}{2}(2(0) + \frac{2}{3}(16-1))$$

$$= 8(0 + 10)$$

$$= \boxed{80}$$

$$\textcircled{34} \sum_{n=3}^{100} -\frac{n}{4} + \frac{3}{4} = 0 - \frac{1}{4} - \frac{2}{4} - \frac{3}{4} - \frac{4}{4} + \dots + \left(-\frac{100}{4} + \frac{3}{4}\right)$$

Arithmetic

$$S_n = \frac{n}{2} (2a_1 + D(n-1))$$

$$S_{98} = \frac{98}{2} (2(0) - \frac{1}{4}(98-1))$$

$$= 49(0 - \frac{97}{4})$$

$$= 49(-\frac{97}{4})$$

$$= \boxed{-1,188.25}$$

Find the value of each infinite geometric series if it exists.

$$\bullet \textcircled{35} \sum_{n=1}^{\infty} \left(\frac{3}{7}\right)^n = \frac{3}{7} + \left(\frac{3}{7}\right)^2 + \left(\frac{3}{7}\right)^3 + \dots$$

$$a_1 = \frac{3}{7} \quad r = \frac{3}{7}$$

$$-1 < r < 1$$

$$-1 < \frac{3}{7} < 1$$



$$S_{\infty} = \frac{a_1}{1-r}$$

$$= \frac{\frac{3}{7}}{1 - \frac{3}{7}}$$

$$= \frac{\frac{3}{7}}{\frac{4}{7}}$$

$$= \boxed{\frac{3}{4}}$$

$$\textcircled{36} \sum_{n=0}^{\infty} 10\left(-\frac{5}{6}\right)^{n+1} = 10\left(-\frac{5}{6}\right) + 10\left(-\frac{5}{6}\right)^2 + 10\left(-\frac{5}{6}\right)^3 + \dots$$

$$a_1 = 10\left(-\frac{5}{6}\right) = -\frac{50}{6} = -\frac{25}{3}, \quad r = -\frac{5}{6}$$

$$-1 < r < 1$$

$$-1 < -\frac{5}{6} < 1$$



$$S_{\infty} = \frac{a_1}{1-r}$$

$$= \frac{-\frac{25}{3}}{1 - \left(-\frac{5}{6}\right)}$$

$$= \frac{-\frac{25}{3}}{\frac{11}{6}}$$

$$= \boxed{\frac{-50}{11}}$$

$$\star \textcircled{37} \sum_{n=2}^{\infty} 2^{n-1} = 2 + 2^2 + 2^3 + \dots$$

$$a_1 = 2, \quad r = 2$$

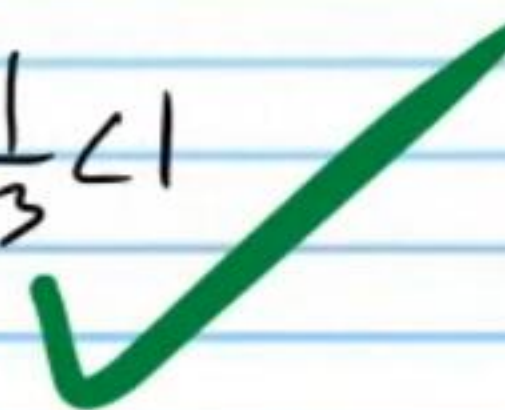
Because $r=2$ is not between -1 and 1 , the sum does not exist.

$$\bullet \textcircled{38} \sum_{n=0}^{\infty} \left(-\frac{1}{3}\right)^n = 1 - \frac{1}{3} + \frac{1}{9} - \frac{1}{27} + \dots$$

$$a_1 = 1, \quad r = -\frac{1}{3}$$

$$-1 < r < 1$$

$$-1 < -\frac{1}{3} < 1$$



$$S_{\infty} = \frac{a_1}{1-r}$$

$$= \frac{1}{1 - \left(-\frac{1}{3}\right)}$$

$$= \frac{1}{1 + \frac{1}{3}}$$

$$= \frac{1}{\frac{4}{3}}$$

$$= \boxed{\frac{3}{4}}$$

Warning! $-0.8^n \neq (-0.8)^n$

$$\textcircled{39} \sum_{n=1}^{\infty} -0.8^n = -0.8 - 0.64 - 0.512 + \dots$$

$$a_1 = -0.8 \quad r = 0.8$$

$$-1 < r < 1$$

$$-1 < 0.8 < 1$$

$$S_{\infty} = \frac{a_1}{1-r}$$

$$= \frac{-0.8}{1-0.8}$$

$$= \frac{-0.8}{0.2}$$

$$= -\frac{8}{2}$$

$$= \boxed{-4}$$

★ $\textcircled{40} \sum_{n=1}^{\infty} (-10)^{n-1} = 1 - 10 + 100 - 1000 + \dots$

Because $r = -10$ is not between -1 and 1 , the sum does not exist.